

ALAMANDA HASITHA

Undergraduate Student at IIIT Hyderabad

📞 9502072613

✉ alamanda.hasitha@students.iiit.ac.in

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Education

International Institute of Information Technology, Hyderabad

Aug 2023 – July 2027

B.Tech in Electronics and Communication Engineering

CGPA: 6.65

Sri Viswa Junior College

2021 – 2023

Class 12 – IPE Percentage: 91.9%

Sri Chaitanya Techno School

Till 2021

Class 10 – SSC Board Percentage: 97.83%

Relevant Coursework

Data Structures and Algorithms | C Programming | Intro to Processor Architecture | VLSI Design | Analog Electronic Circuits | Linear Algebra | Signal Processing | Digital Systems and Microcontrollers | Statistical Methods in AI

Projects

LLM Quantization & Inference Optimization | *Python, PyTorch, bitsandbytes*

[Link](#)

- Engineered a custom Post-Training Quantization (PTQ) pipeline from scratch to compress FP32 model weights to INT8, optimizing memory footprint and inference latency without relying on external quantization libraries.
- Fine-tuned GPT-2 Large for text classification and conducted rigorous performance benchmarking against NF4 and 8-bit quantized models using the bitsandbytes library.

Sparse Mixture-of-Experts (MoE) Architecture | *Python, PyTorch, LLMs*

[Link](#)

- Architected and trained a Sparse MoE Transformer layer from scratch, implementing custom Token Choice Top-k routing algorithms and a custom load balancer to replace standard dense FFN layers.
- Instruction-tuned open-source LLMs (Llama 3, Qwen) for extreme summarization tasks, evaluating generation quality against baseline models using ROUGE and BERTScore metrics.

Rooftop Solar Potential Mapping | *Machine Learning*

[Link](#)

- Developed and optimized a Random Forest model using 7 years of data (5 years training, 2 years testing) to assess rooftop solar potential in the Delhi NCR region.
- Achieved an F1-score of 0.80, demonstrating robust classification performance on real-world geospatial and temporal data.

BUY-SELL App | *MERN*

[Link](#)

- Developed a full-stack e-commerce platform using MERN stack allowing users to create listings, filter items and purchase products. Also implemented secure JWT authentication, user profile management, product CRUD operations and integrated responsive frontend for seamless cross-device experience.

PyBullet Motion-Planning Dataset Generator and Classifier | *Python, PyBullet, Transformers, Robotics*

[Link](#)

- Developed a Python-based robotics data-generation script using PyBullet to create a motion-planning dataset for a planar manipulator.
- Implemented and trained a Transformer model on the generated spatial dataset for motion prediction and achieved a 99.78% accuracy rate.

Technical Skills

Programming Languages: JavaScript, Python, C, C++, HTML, CSS, SQL, Dart

Tools & Frameworks: MongoDB, Node.js, Express, VS Code, MATLAB, Magic, NGSpice, LTSpice, Verilog, Tinkercad, Linux, GitHub, React, Arduino, Flask, FPGA

Achievements & Extracurricular

- Secured an All India Rank of 8,400 (CRL) in JEE Mains 2023 among 1 million+ applicants.
- Volunteered at IIIT Hyderabad's NSS Blood Camp, supporting blood donation drives to promote community health.
- Member in Marketing Team for Felicity – IIIT Hyderabad (Spring 2024).